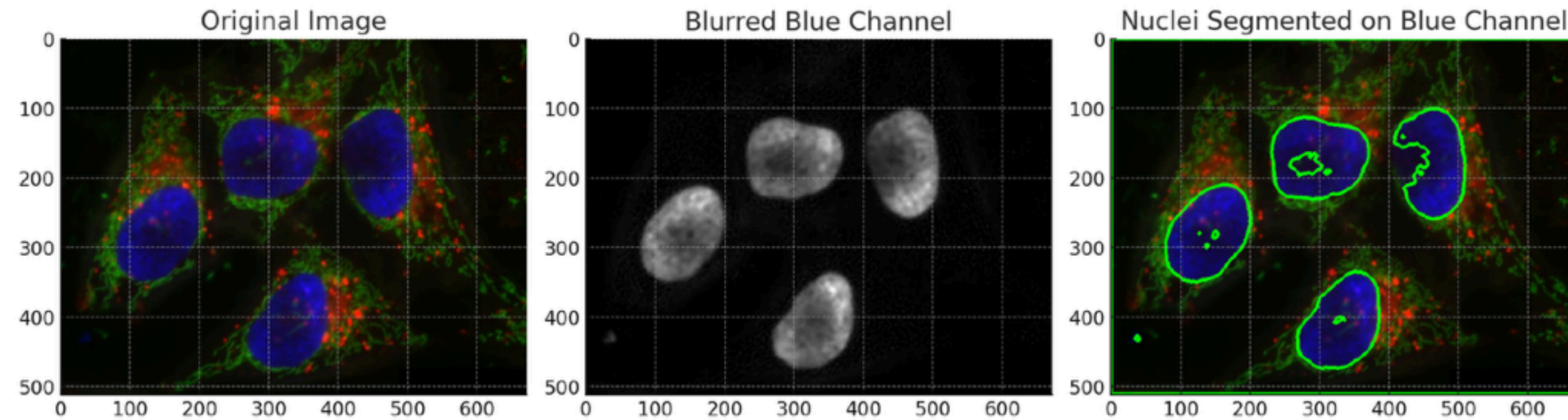


Using ChatGPT-4 for Image Analysis

You
please use the blue channel for segmenting the nuclei

ChatGPT



The nuclei segmentation has been performed using the blue channel of the image with the following steps:

- The blue channel was extracted from the original image since the nuclei appear to be stained blue.
- Gaussian Blur was applied to the blue channel to smooth the image, which helps in reducing noise for thresholding.
- Otsu's thresholding was then used on the blurred blue channel to create a binary image where the nuclei are distinguished from the background.

Using ChatGPT-4 for Image Analysis

Here is the code that was used for this process:

python

 Copy code

```
# Extract the blue channel, as the nuclei are stained with a blue dye
blue_channel = image[:, :, 0]

# Apply GaussianBlur to the blue channel
blurred_blue = cv2.GaussianBlur(blue_channel, (5, 5), 0)

# Perform Otsu's thresholding on the blue channel
_, blue_thresh = cv2.threshold(blurred_blue, 0, 255, cv2.THRESH_BINARY_

# Find contours on the thresholded blue channel
contours_blue, _ = cv2.findContours(blue_thresh, cv2.RETR_TREE, cv2.CHA

# Draw the contours on the original image to show the segmentation of n
contoured_blue_image = image.copy()
cv2.drawContours(contoured_blue_image, contours_blue, -1, (0, 255, 0),
```